

Fraud Detection in Applications

Internee.pk Data Analyst Internship - Task 4 (Major Project)
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Applications	Suspicious	High/Critical	Duplicate	Rapid	Inconsistent	IF Flags
720	121	117	80	39	20	80

Objective

Identify anomalies in internship applications to prevent fake or repeated entries. The project analyzes duplicate patterns, rapid submissions, and inconsistent data, then applies Isolation Forest and K-Means Clustering to prioritize suspicious applications for human review.

Responsible-use principle: a suspicious score or model flag is not proof of fraud and must not be used as an automatic rejection decision.

Methodology and Task 4 Coverage

Requirement	Implementation
Duplicate entries	Repeated email, phone and CV fingerprints plus duplicate group counts.
Rapid submissions	Submission gaps, device/IP 24-hour activity and very fast form completion.
Inconsistent data	CGPA/scale, semester/graduation timing, availability and short-form contradictions.
Isolation Forest	350-tree unsupervised model on 16 standardized engineered features with 11% contamination.
K-Means	4-cluster behavior segmentation with 25 initializations.
Suspicious alerts	Combined risk score, alert level, explainable reason and interactive review queue.

Model Validation on Synthetic Reference Scenarios

Metric	Result
Synthetic anomalies planted	80
Isolation Forest flagged	80
High/Critical queue	117
Precision	65.81%
Recall	96.25%
F1	78.17%

The reference label is excluded from unsupervised model fitting and is used only to check whether the portfolio experiment surfaces deliberately planted anomaly scenarios.

Key Findings and Manager Recommendations

1. Duplicate behavior is a major review signal

80 of 720 applications contain a duplicate email, phone or CV pattern in the full synthetic dataset.

2. Rapid behavior is visible in burst patterns

39 applications meet at least one rapid-submission condition, including very short form completion or repeated device/IP activity.

3. Inconsistency checks add explainability

20 records trigger one or more internal consistency rules.

4. Isolation Forest adds multivariate anomaly detection

80 applications are flagged by the unsupervised model, including cases that may not be obvious from a single rule.

5. Human review remains essential

Use model and rule outputs to prioritize review. Verify evidence before accepting, rejecting or escalating an application.

Recommended Controls

- Validate and deduplicate email, phone and CV fingerprints during submission.
- Apply sensible rate limits and review repeated rapid activity by device and hashed network identifier.
- Validate academic and availability fields at data-entry time to reduce contradictions.
- Prioritize Critical and High alerts first, then review a sample of medium-risk cases.
- Re-evaluate model thresholds and cluster behavior as application patterns change.
- Do not use location, university or other contextual attributes as automatic fraud indicators.

Deliverables

Interactive HTML dashboard, local CSS/JavaScript, 720-row CSV, formatted Excel audit workbook, data dictionary, Python generation and ML validation scripts, README, submission guide, checklist, LinkedIn post draft, demo-video script, Windows launcher, dashboard screenshot, and this analysis summary.